Code No: 115AP **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, September - 2021
COMPILER DESIGN
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks

1.a) List out different phases of compiler and explain each phase role in compilation process
b) Explain the bootstrapping. [8+7]

2.a) Write a note on complier construction tools.
b) Construct recursive descendent parsing table for following grammar. [7+8]
E→E+T/T
T→TF/F
F→F*/a/b

3.a) What is LALR parser? Explain LALR parsing algorithm.
b) Write a note on YACC**.** [8+7]

4.a) What is shift reduce parser? Explain conflicts in shift reduce parser.
b) Write LR parsing algorithm. [7+8]

5.a) Explain various different intermediate code forms.
b) Write intermediate code for assignment statement and case statement. [7+8]

6.a) What is syntax directed translation schem and write syntax directed schem for **while** statement.
b) Explain different formants of symbol table. [8+7]

7.a) Explain, how dead code elimination is done and illustrate with example.
b) Explain optimization of basic blocks. [7+8]

8.a) Write the issues in design of code generation.
b) Write the steps for constructing DAG for register allocation. [7+8]

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